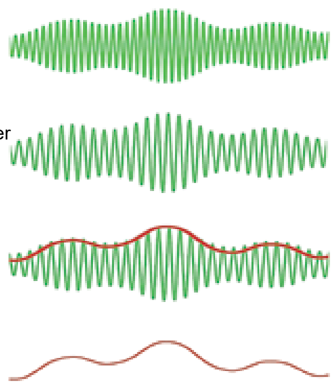

Fig. 3: At receiver side, the broadcasted modulated signal is received, an IF signal is generated using a local oscillator and a signal mixer, and the original audio signal is recovered using a demodulator (or envelope detector)

TABLE 1: VARIOUS VALUES OF F1 AND F2

Carrier (f1)	LO (f2)	Mixer (f2+f1)	Mixer (f2-f1)
600	700	1300	100
800	900	1700	100
1300	1400	2700	100
1400	1500	2900	100
1559	1659	3218	100
1600	1700	3300	100

the weak output of LO is fine for our work here.

Let us also imagine we have a signal mixer that will produce the addition and subtraction of the two signals we provide. Let us say the received modulated carrier frequency is f1 and output of the local oscillator is f2. The mixer will take the signal of frequency f1 and signal of frequency f2, then it will produce addition and subtraction of f1 and f2 and produce its output as f2 + f1, and f2-f1.

We can see an example with various values of f1 and f2 in Table 1.

Resulting outputs f2 + f1 and f2-f1 of the mixer are result of both f1 and f2, and hence both f2 + f1 and f2-f1 contain the information of the modulated carrier wave f1 and LO wave f2.

As you can see, the f2 + f1 keeps changing depending on the f1 and f2, while the f2-f1 is always constant! That is, it remains 100kHz in this example. It means, if we apply a filter on the output of the mixer to select only f2-f1 (that is 100kHz in this example) and filter out all other frequencies, including f2 + f1, then we will end up only with f2-f1 (without the f2 + f1 content).

In other words, to extract any particular station from a bunch of stations, all we need to do is change the LO frequency (f2) such that the f2-f1 is 100kHz!

Thus, for any f1 (of station of our interest), we just need to adjust local oscillator to produce a frequency f2 in such a way that f2-f1 will result in 100kHz. For example, if we want to hear the radio station Akashvani Mumbai, which is transmitting at 558kHz (let us take it as f1), all we now need to do is just adjust the local oscillator to produce 658kHz (let us take it as f2), so that the mixer will produce f2 + f1 and f2-f1. Thereafter, when the mixer output is passed through a filter to allow only f2-f1, we will get f2-f1, which is at 100kHz. The Important thing to note here is, that the 100kHz is the intermediate frequency or IF. When this IF (see Fig. 3) is sent to an envelope detector or demodulator (see Fig. 4), it will remove the carrier wave and give the original audio (baseband signal) out. This audio signal can be further amplified suitably and heard through a loudspeaker.

Did you notice the advantage of the IF? As you can see, we just need an audio amplifier that will amplify only the audio signal. One can also have an amplifier before the envelope detector so that the mixer output is amplified a bit more before sending it to the envelope detector, if needed. The main advantage of the IF is, we can convert the incoming carrier frequency (f1) to a smaller frequency (f2-f1), which is the IF, and it becomes lot more convenient to design all the circuitry after the mixer to work with a fixed frequency.

Why IF is fixed to 455kHz in AM radio

There is a very interesting background for this. Following are some reasons for choosing the exact value of 455kHz for IF:

1. Keep it outside the AM band. It was important to ensure that any radio transmission was not exactly like IF. That is, the IF should be outside the AM band of 531kHz to 1602kHz. That means the IF should be either lower than 531kHz or higher than 1602kHz.

2. Be as less as possible. Lower frequency was better as we did not have to use the expensive circuitry such as filters and IF amplifiers for

Advantages of IF

While IF is not mandatory, here are some advantages of using it:

Consistent performance. Since the IF is fixed, regardless of the actual broadcast frequency, the filters and amplifiers can be optimised for this specific frequency, ensuring consistent performance.

Simplified tuning. It allows the use of a single, fixed-frequency filter and amplifier stages, simplifying the design and tuning of the receiver. In the early days, high frequency and wide-band amplifiers were a lot more expensive than they are now.

Improved selectivity and sensitivity. With a stable IF, the receiver can more effectively select the desired signal and reject others, improving both selectivity and sensitivity.